



ExplainCheck — Publication-First Research & Engineering Protocol



Purpose: This is the authoritative, publication-first specification for building **ExplainCheck**, an open-source benchmark and model-card generator for evaluating explainable AI methods. Treat it as the research protocol, engineering contract, experiment plan, manuscript scaffold, and authorship/accountability record. The software exists to answer the research questions; features that do not strengthen validity, reproducibility, or publication quality are secondary.

Document control

Field	Value
Working project name	ExplainCheck
Primary researcher and accountable author	Saranraj U
AI research and engineering assistant	IGRIS / Notion AI
Study type	Reproducible computational benchmark and open-source research software
Initial modality	Tabular supervised classification
Primary artifact	Peer-reviewed research paper supported by an open-source benchmark
Secondary artifacts	Versioned Python package, benchmark dataset manifests, model cards, JSON results, CI integration
Protocol status	Pre-implementation research protocol
Protocol version	0.1.0
Change rule	Any change to hypotheses, datasets, metrics, exclusions, or analysis after experiments begin must be logged and labelled exploratory

1. Working title

ExplainCheck: A Reproducible, Stress-Tested Benchmark for Evaluating Feature-Attribution Explanations and Generating CI-Ready Model Cards

Alternative titles

- Beyond Visual Plausibility: Benchmarking the Fidelity, Stability, and Stress Robustness of Tabular XAI Methods**
- Do Explanations Survive Data Imperfections? A Benchmark of Feature Attribution under Correlation and Missingness**
- From Explanation to Evidence: Reproducible Quality Gates and Model Cards for Explainable AI**

One-sentence contribution claim

ExplainCheck evaluates local feature-attribution methods under controlled and realistic stressors—particularly correlated features and missing data—quantifies uncertainty and subgroup disparities, and converts the results into reproducible model cards and CI/CD quality gates.

2. Publication-ready abstract template

Background: Post-hoc explanation methods are commonly assessed through visually persuasive examples, despite evidence that explanation quality varies across metrics, models, datasets, and implementation choices. Existing toolkits provide valuable evaluation metrics, but gaps remain in stress-testing tabular explanations under correlated features and missing data and in translating benchmark evidence into deployable quality gates and versioned model documentation.

Objective: We introduce ExplainCheck, a reproducible benchmark for evaluating local feature-attribution explanations across fidelity, stability, sparsity, computational efficiency, correlation robustness, missingness sensitivity, and subgroup consistency.

Methods: We evaluate [EXPLAINER COUNT] explanation methods on [DATASET COUNT] synthetic and real-world tabular datasets and [MODEL COUNT] model families. Synthetic datasets provide known ground-truth feature relevance, while real-world datasets test ecological validity. Correlation and missingness stressors are introduced at prespecified intensities. Metrics are estimated over repeated seeds, accompanied by bootstrap confidence intervals and multiplicity-adjusted statistical comparisons. Evaluation is conditioned on prediction preservation where required to avoid conflating model sensitivity with explanation instability.

Results: [Insert principal findings, effect sizes, confidence intervals, and ranking changes. Do not write this section before results are frozen.]

Conclusion: [Insert evidence-supported conclusion.] ExplainCheck releases machine-readable benchmark artifacts, publication-ready tables and figures, and model cards suitable for continuous integration, enabling explanation quality to be treated as a testable property rather than a visual assertion.

Keywords: explainable artificial intelligence; feature attribution; benchmark; fidelity; stability; missing data; correlated features; model cards; reproducibility; MLOps

3. Research problem and motivation

Explainability is frequently presented as a visualization rather than evaluated as an empirical claim. SHAP assigns feature contributions through a unified additive-attribution framework, while LIME fits an interpretable surrogate near an individual prediction. Their explanations are useful only to the extent that they are faithful to the model, stable under semantically insignificant changes, concise enough to interpret, computationally feasible, and robust to data characteristics such as collinearity and missingness.

The literature already establishes that one metric is insufficient. Quantus provides a broad collection of quantitative XAI metrics, and OpenXAI benchmarks post-hoc explanation methods across faithfulness, stability, and fairness. However, published surveys report limited standardization and inconsistent coverage across toolkits. ExplainCheck must therefore **not** claim that XAI benchmarking itself is novel. Its defensible contribution is the combination of:

1. A controlled **tabular stress-test protocol** for correlation and missingness.
2. Metric computation that explicitly separates changes in the predictive model from failures of the explanation method.
3. Uncertainty-aware, subgroup-aware reporting instead of unqualified point estimates.
4. Reproducible artifacts that connect experiments to **versioned model cards and CI quality gates**.
5. A transparent benchmark manifest that records datasets, transformations, explainers, parameters, seeds, dependencies, hardware, exclusions, and hashes.

Model cards motivate standardized documentation of intended use, evaluation conditions, subgroup performance, limitations, and ethical considerations. ExplainCheck extends this reporting idea by automatically inserting evidence about explanation behavior. This aligns with broader trustworthy-AI principles emphasizing validity, reliability, transparency, explainability, and fairness.

4. Research gap

Established by prior work

- Visual inspection alone is insufficient for comparing explanations.
- Explanation metrics can disagree, and rankings may depend on metric definitions and parameters.
- Existing frameworks such as Quantus and OpenXAI provide important evaluation foundations.
- Model cards improve transparency by documenting intended uses, evaluation conditions, limitations, and subgroup behavior.
- Missing-data handling can affect predictions and model interpretation.

Gap targeted by this study

There is insufficient standardized evidence about whether common tabular feature-attribution explanations remain reliable when:

- predictive performance remains approximately stable but the feature correlation structure changes;
- missing values are introduced under controlled mechanisms and handled using different strategies;
- explanation quality is examined across subgroups rather than only in aggregate;
- a model is updated and explanation regressions must be detected automatically;
- metric results must be converted into auditable, machine-readable deployment documentation.

Explicit non-novelty statement

ExplainCheck is not the first XAI evaluation toolkit, model-card generator, SHAP/LIME comparison, or robustness benchmark. Publication viability depends on rigorously demonstrating a new stress-testing protocol, new empirical findings, and a reproducible bridge from XAI evaluation to continuous model governance.

5. Research objectives

Primary objective

Develop and empirically validate a reproducible benchmark that measures how explanation methods behave across models and datasets under correlation and missingness stressors.

Secondary objectives

- Determine whether explainer rankings change across evaluation criteria.
 - Quantify the extent to which missingness and correlation alter explanations without materially changing predictions.
 - Assess whether explanation quality differs across prespecified subgroups.
 - Evaluate the repeatability and computational cost of explanation methods.
 - Generate versioned model cards and JSON artifacts directly from frozen experiment results.
 - Demonstrate CI/CD checks that identify explanation-quality regressions between model versions.
-

6. Research questions and hypotheses

RQ1 — Metric dependence

Question: Do explanation methods receive consistent rankings across fidelity, stability, sparsity, and runtime?

H1: No single explanation method will dominate across all criteria; rankings will show statistically and practically meaningful disagreement.

RQ2 — Model and dataset dependence

Question: How strongly do model family and dataset characteristics affect explanation-method performance?

H2: Explainer performance will exhibit significant explainer × model and explainer × dataset interactions.

RQ3 — Correlation robustness

Question: Do explanations remain stable when redundant or correlated features are introduced while predictive behavior is preserved?

H3: Feature-level rankings will be unstable under strong correlation, while grouped attribution measures will be more robust than ungrouped measures.

RQ4 — Missingness sensitivity

Question: How do missingness rate, mechanism, and handling strategy affect predictions and explanations?

H4: Attribution drift will increase with missingness rate and will differ by imputation strategy, even among samples whose predicted class remains unchanged.

RQ5 — Subgroup consistency

Question: Are fidelity and stability comparable across prespecified subgroups?

H5: Aggregate metrics will conceal subgroup differences in at least some dataset–model–explainer configurations.

RQ6 — Explanation regression testing

Question: Can prespecified CI quality gates detect explanation regressions that ordinary predictive-performance checks miss?

H6: At least one controlled model update will preserve predictive performance within tolerance while violating a prespecified explanation-quality threshold.

Exploratory question

How correlated are the benchmark metrics, and which subset provides non-redundant information? This analysis must remain exploratory unless metric selection is frozen before experiments.

7. Scope and non-goals

Included in the primary paper

- Tabular binary classification
- Local feature attribution
- Synthetic ground-truth datasets and public real-world datasets
- Logistic Regression, Random Forest, and gradient-boosted trees
- KernelSHAP, TreeSHAP where compatible, and LIME
- Fidelity, stability, sparsity, runtime, correlation robustness, missingness sensitivity, and subgroup consistency
- Repeated experiments with uncertainty estimates
- Model-card and JSON generation
- CI regression demonstration

Secondary or appendix-only

- Global permutation importance and mean absolute SHAP, evaluated in a separate global-explanation track
- Counterfactual validity as an optional extension
- HTML dashboard
- Additional model families

Excluded from the first paper

- Image, text, and time-series explanations
- Human-subject trust or comprehension studies
- Causal interpretation claims
- Claims that explanations establish model fairness
- A universal weighted “XAI score”
- Production deployment in a high-stakes setting



Critical validity rule: Local and global explanations must not be placed in the same leaderboard. Permutation importance is a global method; SHAP/LIME can produce local explanations. Maintain separate tracks and compare only methods answering the same explanatory question.

8. Conceptual definitions

- **Explanation:** A method-produced representation intended to characterize aspects of a trained model's behavior for a specified input and target output.
- **Feature attribution:** A vector assigning signed or unsigned relevance to input features.
- **Fidelity/faithfulness:** The degree to which an explanation reflects the behavior of the predictive model under a defined intervention or perturbation protocol.
- **Stability:** Similarity of explanations for semantically similar inputs, evaluated only when the model output remains within a prespecified invariance condition.
- **Sparsity:** The number or proportion of features required to represent a prespecified share of attribution magnitude.
- **Robustness to correlation:** Resistance of an explanation's substantive conclusion to redundant or correlated representations of the same signal.
- **Missingness sensitivity:** Change in predictions or explanations after controlled missingness and handling procedures.
- **Subgroup consistency:** Difference in the distribution of explanation-quality metrics across prespecified groups. It is a diagnostic and not proof of fairness.
- **Counterfactual validity:** Proportion of generated counterfactuals that achieve the target prediction while satisfying domain and actionability constraints.

9. Experimental design

9.1 Unit of analysis

The primary unit is an explanation for one sample, target class, trained model, explainer, dataset condition, and random seed. Aggregated comparisons must account for repeated measurements and dataset clustering.

9.2 Dataset strategy

Use two complementary groups.

A. Synthetic datasets with ground truth

Generate datasets where the true predictive features and functional relationships are known. Include:

1. Linear additive signal
2. Non-linear interactions
3. Redundant correlated features
4. Irrelevant noise features
5. Controlled subgroup-dependent feature distributions
6. Controlled MCAR and MAR missingness; treat MNAR as exploratory because it depends on unobserved values and strong assumptions

For every generator, save:

- source code and version;
- generating equation;
- parameter values;
- true relevant feature set;
- seed;
- sample size and dimensionality;
- correlation matrix;
- missingness mechanism and rate.

B. Real-world public datasets

Select three to five tabular binary-classification datasets with different sizes, dimensionalities, imbalance levels, and domain contexts. Candidate classes include census income, credit risk, bank marketing, medical prediction, and electricity or sensor data.

Dataset inclusion criteria

- Publicly accessible with a redistribution-compatible license
- Stable version or immutable snapshot
- Documented target and feature semantics
- At least one dataset with meaningful subgroup attributes
- Feasible on commodity hardware
- No requirement to expose personal or restricted data

Before final selection: verify license, provenance, preprocessing conventions, known ethical issues, subgroup meaning, and whether the dataset is appropriate for the claims. Create a datasheet for every dataset. Do not choose datasets merely because earlier XAI papers used them.

9.3 Data splitting

- Freeze a train/validation/test split per seed.
- Use stratified splitting for classification.
- Fit preprocessors only on training data.

- Fit explainer background/reference data only from training data.
- Evaluate explanations only on the held-out test set.
- Preserve identical splits across explainers within each run.
- Save row identifiers and split hashes.

9.4 Model families

Model	Purpose	Expected explanation method
Logistic Regression	Transparent linear baseline	Coefficients plus KernelSHAP/LIME as post-hoc checks
Random Forest	Non-linear ensemble	TreeSHAP, KernelSHAP, LIME
XGBoost or LightGBM	High-performing boosted trees	TreeSHAP, KernelSHAP, LIME

Optional: a compact multilayer perceptron only if it adds a clear research contribution and does not compromise reproducibility.

Tune models using validation data or nested cross-validation. Freeze search spaces in configuration. Report predictive discrimination, calibration, and class imbalance metrics before interpreting explanation results.

9.5 Explanation methods

- **TreeSHAP:** model-specific attribution for tree ensembles.
- **KernelSHAP:** model-agnostic attribution with recorded background set and sampling budget.
- **LIME:** local surrogate with recorded kernel width, discretization, neighborhood sampling, feature count, and seed.
- **Transparent reference:** logistic-regression coefficients or known synthetic ground truth where applicable.

Do not claim that one explainer is intrinsically better based on one metric. Record every explainer parameter because rankings can be parameter-sensitive.

9.6 Sample size and repetitions

Initial target:

- 3–5 real datasets
- 4–6 synthetic scenarios
- 3 model families
- 3 local explainers where compatible
- 10 independent training seeds
- At least 200 held-out explanation samples per dataset/seed, increased after runtime pilot
- 20 local perturbations per sample for stability, increased if Monte Carlo error remains excessive
- 1,000 stratified bootstrap replicates for confidence intervals

Conduct a pilot to estimate variance and compute budget. Freeze final sample counts before the confirmatory run. If computational constraints require reduction, document the decision before seeing final comparative results.

10. Metric specification

Every metric implementation must state its target construct, formula, direction, range, assumptions, perturbation distribution, aggregation method, and known failure modes.

10.1 Fidelity

Use complementary measures rather than a single universal proxy.

Perturbation fidelity / deletion impact

For attribution vector $\mathbf{a}$ and top- k feature set S_k , perturb features in ranked order and measure the change in the model score:

$$AOPC = (1/K) \times \sum_{k=1..K} (f(x) - f(x \text{ perturbed on } S_k))$$

A larger score indicates that highly ranked features materially influence the model, but results depend on the perturbation baseline. Use data-aware conditional or empirical replacements when possible and report sensitivity to the replacement strategy.

Local surrogate fidelity

For LIME-like surrogates, report weighted local R^2 or weighted prediction agreement within the sampled neighborhood. Do not use this metric to advantage methods without an explicit surrogate; retain it as a method-specific diagnostic.

Synthetic ground-truth recovery

Where true relevant features are known, report top- k precision, recall, F1, and rank correlation against known relevance.

10.2 Stability

Generate valid local perturbations δ such that the model prediction is preserved:

$$I[f(x + \delta) = f(x)] = 1$$

Then compare attribution vectors using:

- cosine similarity;
- Spearman rank correlation;
- top- k Jaccard overlap.

Report the three components separately. Do not collapse them into one score in the primary analysis.

$$\text{Jaccard@k} = |\text{Top}_k(\mathbf{a}(x)) \cap \text{Top}_k(\mathbf{a}(x + \delta))| / |\text{Top}_k(\mathbf{a}(x)) \cup \text{Top}_k(\mathbf{a}(x + \delta))|$$

Also report the fraction of candidate perturbations rejected because they changed the model prediction. This prevents explanation instability from being confused with genuine model sensitivity.

10.3 Sparsity

Define attribution-mass sparsity as the smallest number of features required to explain τ of total absolute attribution:

$$k\tau = \min k \text{ such that } \sum_{i=1..k} |a(i)| / \sum_{j=1..d} |a(j)| \geq \tau$$

Use $\tau = 0.90$ as the primary threshold and $\tau \in \{0.80, 0.95\}$ for sensitivity analysis. Report both $k\tau$ and $k\tau/d$.

10.4 Runtime and resource use

Report separately:

- explainer initialization time;

- per-sample explanation latency;
- P50, P95, and P99 latency;
- peak memory;
- throughput;
- model inference time;
- hardware, threads, and parallelization.

Run timed comparisons after warm-up and in randomized method order. Runtime conclusions are hardware- and implementation-dependent.

10.5 Correlation robustness

Create controlled redundant features with target correlations $\rho \in \{0.3, 0.6, 0.9\}$. Compare baseline and stressed conditions while monitoring predictive invariance.

Primary outputs:

- top- k rank overlap;
- attribution-vector similarity after aligned feature mapping;
- attribution mass assigned to the original-plus-redundant feature group;
- change in prediction;
- group-attribution conservation.

$$\text{GroupConservation}(g) = 1 - |A_{\text{baseline}}(g) - A_{\text{stressed}}(g)| / (|A_{\text{baseline}}(g)| + \epsilon)$$

Interpret feature-level instability cautiously: redistribution among equivalent correlated features may not imply that the group-level explanation is wrong.

10.6 Missingness sensitivity

Stress rates: 5%, 10%, 20%, and 30%.

Primary mechanisms:

- MCAR
- MAR generated from observed covariates

Exploratory mechanism:

- Simulated MNAR, accompanied by explicit assumptions

Handling strategies:

- median/mode imputation;
- KNN imputation;
- iterative imputation or MICE-style approach;
- native missing-value handling when supported;
- missingness-indicator augmentation.

Measure:

- prediction drift;
- class-flip rate;
- attribution cosine similarity;
- top- k Jaccard overlap;

- fidelity change;
- subgroup-specific changes.

Primary explanation-drift analysis must use samples with prediction preservation. Report class-flip cases separately as joint model-and-explanation sensitivity.

10.7 Subgroup consistency

For each prespecified group g , calculate metric distributions and compare them using:

$$\text{Gap}(m, g1, g2) = | E[m | g1] - E[m | g2] |$$

Also report standardized effect sizes and confidence intervals. Where sample sizes permit, evaluate intersectional groups. Do not infer discriminatory intent or causal unfairness from metric gaps.

10.8 Counterfactual validity — optional extension

If included, evaluate validity, proximity, sparsity, plausibility, actionability, constraint violations, and diversity. Keep counterfactual evaluation separate from feature-attribution leaderboards.

```
ValidityRate = valid counterfactuals / requested counterfactuals
Proximity = normalized L1 or mixed-type distance
Sparsity = number of changed features
ConstraintViolationRate = invalid changes / generated counterfactuals
```

11. Statistical analysis plan

Confirmatory analysis

1. Report mean, median, standard deviation, and 95% confidence intervals for every primary metric.
2. Use paired comparisons because explainers operate on the same dataset/model/sample conditions.
3. Prefer hierarchical mixed-effects models where feasible:

```
metric ~ explainer × model × stress_level + (1 | dataset) + (1 | seed)
```

1. If distributional assumptions are poor, use aligned-rank or Friedman tests within comparable blocks, followed by multiplicity-adjusted paired post-hoc tests.
2. Correct families of comparisons using Holm's procedure.
3. Report effect sizes, not only p -values.
4. Use stratified bootstrap confidence intervals at the dataset or run level rather than treating every explanation as independent.
5. Distinguish statistical significance from a prespecified practical threshold.

Ranking analysis

- Compute per-dataset and aggregate ranks.
- Report rank uncertainty.
- Measure agreement between metrics using Kendall's τ .
- Never hide trade-offs behind a single weighted score.
- If a composite score is explored, publish all weights and show sensitivity to weight choices; label it exploratory.

Missing data in experimental outputs

Failed explainer runs are results, not observations to silently delete. Record failure class, stack trace hash, stage, dataset, model, and parameters. Report failure rates by method. Define retry rules before the final run.

Exclusion policy

Permitted exclusions:

- corrupted artifact;
- deterministic implementation failure confirmed by a minimal reproducer;
- violation of a prespecified numerical-validity check.

Every exclusion must be logged. Outliers may be visualized and analyzed robustly but should not be removed merely because they weaken a hypothesis.

12. Benchmark quality gates

Quality gates are contextual and must not be advertised as universal scientific standards.

Example configuration:

```
project:
  name: explaincheck-study
  protocol_version: 0.1.0

reproducibility:
  seeds: [11, 23, 37, 41, 53, 67, 71, 83, 97, 101]
  deterministic: true

metrics:
  fidelity:
    enabled: true
    practical_regression_tolerance: 0.03
  stability:
    enabled: true
    min_median_jaccard_at_5: 0.70
  sparsity:
    enabled: true
    attribution_mass: 0.90
  runtime:
    enabled: true
    max_p95_ms: 500
  missingness_sensitivity:
    rates: [0.05, 0.10, 0.20, 0.30]
    max_median_drift_at_10pct: 0.20

ci:
  compare_against: artifacts/baseline/benchmark.json
  fail_on_regression: true
```

A CI failure means “this model violated the project’s declared tolerance,” not “the model is universally unexplainable.”

13. Software architecture

```

explaincheck/
├── pyproject.toml
├── README.md
├── LICENSE
├── CITATION.cff
├── CODE_OF_CONDUCT.md
├── CONTRIBUTING.md
├── configs/
│   ├── protocol-v0.1.yaml
│   └── smoke-test.yaml
├── src/explaincheck/
│   ├── contracts/
│   ├── datasets/
│   ├── models/
│   ├── explainers/
│   ├── metrics/
│   ├── perturbations/
│   ├── stressors/
│   ├── statistics/
│   ├── reporting/
│   ├── model_cards/
│   ├── cli/
│   └── provenance/
├── experiments/
│   ├── manifests/
│   ├── runners/
│   └── slurm-or-local/
├── tests/
│   ├── unit/
│   ├── property/
│   ├── integration/
│   └── golden/
├── paper/
│   ├── manuscript.tex
│   ├── references.bib
│   ├── tables/
│   └── figures/
├── artifacts/
│   ├── raw/
│   ├── tidy/
│   ├── reports/
│   └── model-cards/
├── .github/workflows/
│   ├── test.yml
│   ├── benchmark-smoke.yml
│   └── explanation-regression.yml

```

Architectural principles

- Experiment configuration is immutable and hashable.
- Metrics are plugins with explicit contracts.
- Explainers return a common typed representation.
- Raw results are append-only.
- Derived tables and figures are regenerated from raw artifacts.
- Manuscript numbers must come from generated files, not manual transcription.
- Every run records code commit, environment lock, dataset hashes, model hash, config hash, seed, and hardware.

Core interfaces

```

class Explainer(Protocol):
    name: str
    scope: Literal["local", "global"]

    def fit(self, model, background_data, *, seed: int) -> None: ...

```

```
def explain(self, X, *, target) -> "ExplanationBatch": ...

class Metric(Protocol):
    name: str
    direction: Literal["higher", "lower"]

    def evaluate(
        self,
        model,
        explainer,
        explanations,
        dataset_context,
        *,
        seed: int,
    ) -> "MetricResult": ...
```

Explanation contract

```
{
  "schemaVersion": "1.0.0",
  "scope": "local",
  "sampleId": "test-00421",
  "target": 1,
  "featureNames": ["age", "income", "utilization"],
  "attributions": [0.12, -0.43, 0.31],
  "baseValue": 0.28,
  "prediction": 0.74,
  "explainer": {
    "name": "kernelshap",
    "version": "x.y.z",
    "parameters": {}
  }
}
```

14. Required research outputs

The pipeline must produce artifacts that can be inserted directly into a paper.

```
artifacts/run-<id>/
├─ run-manifest.json
├─ environment.txt
├─ raw-results.parquet
├─ tidy-results.csv
├─ failures.csv
├─ statistical-tests.csv
├─ benchmark.json
├─ model-card.md
├─ model-card.html
├─ methods-generated.md
├─ limitations-generated.md
├─ tables/
│   └─ table-1-datasets.csv
│   └─ table-1-datasets.tex
│   └─ table-2-model-performance.csv
│   └─ table-2-model-performance.tex
│   └─ table-3-xai-metrics.tex
│   └─ table-4-stress-tests.tex
```

```

├── table-s1-full-results.tex
├── figures/
│   ├── fig-1-study-design.pdf
│   ├── fig-2-metric-tradeoffs.pdf
│   ├── fig-3-correlation-robustness.pdf
│   ├── fig-4-missingness-sensitivity.pdf
│   ├── fig-5-subgroup-gaps.pdf
│   └── fig-s1-runtime.pdf
├── paper-snippets/
│   ├── results-summary.tex
│   ├── key-numbers.json
│   └── captions.md

```

Tidy results schema

One row per metric observation:

Column	Description
protocol_version	Frozen protocol identifier
run_id	Unique run
dataset_id	Versioned dataset identifier
split_hash	Train/validation/test fingerprint
model_family	Model class
model_hash	Serialized model fingerprint
explainer	Explanation method
explainer_scope	Local or global
metric	Metric name
metric_variant	Exact operationalization
stressor	None, correlation, or missingness
stress_level	Numeric level
subgroup	Prespecified group or <code>all</code>
seed	Random seed
sample_id	Test sample identifier, if local
value	Raw metric value
prediction_preserved	Boolean
runtime_ms	Measured time
status	success, failed, excluded

Paper-number rule

The manuscript must reference macros or generated tables:

```

\newcommand{\MedianStabilitySHAP}{0.82}
\newcommand{\MedianStabilityLIME}{0.67}

```

Never manually copy a number from a notebook into the final manuscript.

15. Versioned benchmark JSON

```

{
  "schemaVersion": "1.0.0",
  "protocolVersion": "0.1.0",
  "project": {
    "name": "ExplainCheck",

```

```

    "studyId": "EC-TABULAR-001"
  },
  "provenance": {
    "gitCommit": "<sha>",
    "configurationHash": "sha256:<hash>",
    "environmentHash": "sha256:<hash>",
    "hardware": "<description>",
    "startedAt": "<ISO-8601>",
    "completedAt": "<ISO-8601>"
  },
  "dataset": {
    "id": "<dataset-version>",
    "hash": "sha256:<hash>",
    "datasheet": "<path>"
  },
  "model": {
    "family": "xgboost",
    "version": "1.2.0",
    "hash": "sha256:<hash>",
    "performance": {}
  },
  "results": [
    {
      "explainer": "treeshap",
      "metric": "stability.topKJaccard",
      "estimate": 0.82,
      "ci95": [0.79, 0.85],
      "n": 2000,
      "direction": "higher",
      "threshold": 0.70,
      "status": "passed"
    }
  ],
  "limitations": [],
  "overallStatus": "passed"
}

```

Validate this artifact against a published JSON Schema. Increment `schemaVersion` only for contract changes and `protocolVersion` for scientific-design changes.

16. Model card template

Model details

- Model name and version
- Developer and accountable owner
- Date, license, model family, training procedure
- Code, model, dataset, and configuration hashes

Intended use

- Intended users
- Supported decision context

- Out-of-scope and prohibited uses
- Required human oversight

Data

- Training, validation, and evaluation dataset versions
- Datasheet links
- Missingness and subgroup composition
- Preprocessing and feature handling

Predictive performance

- Discrimination
- Calibration
- Error profile
- Subgroup performance
- Confidence intervals

Explanation methods

- Methods and versions
- Background/reference data
- Parameters
- Local/global scope
- Target output explained

Explanation-quality evidence

- Fidelity
- Stability
- Sparsity
- Runtime
- Correlation robustness
- Missingness sensitivity
- Subgroup consistency
- Failed checks and uncertainty

Limitations and ethical considerations

- Explanations describe model behavior, not causality
- Feature correlation may redistribute attribution
- Missing-data conclusions depend on simulated mechanisms
- Subgroup gaps are diagnostics, not complete fairness assessments
- Quality thresholds are context-specific

Reproduction

```
explaincheck reproduce --manifest artifacts/run-<id>/run-manifest.json
```


17. Testing strategy

Unit tests

- Metric formulas on hand-computed fixtures
- Attribution normalization
- Seed propagation
- Schema validation
- Hash stability
- Missingness generator invariants

Property-based tests

- Identical inputs and seeds produce identical explanations when the method is deterministic
- Attribution dimensions always match the feature schema
- Missingness rates remain within tolerance
- Group attribution equals the sum of mapped member attributions
- Metric direction and range constraints hold

Integration tests

- Dataset → model → explanation → metric → report
- All supported explainer/model combinations
- CI baseline comparison
- Resume after interrupted runs

Golden scientific tests

- Linear synthetic model recovers known feature ranking
- Noise features receive near-zero importance within tolerance
- Duplicated correlated features conserve group attribution better than individual attribution
- Increasing perturbation intensity produces a detectable stability response

Independent verification

Before submission, manually reproduce at least one full experiment from a clean environment using only the public instructions. Ideally, have another person run the replication or use a fresh cloud runner.

18. Reproducibility and open-science protocol

- Freeze dependencies with a lockfile.
- Publish a container definition.
- Use deterministic seeds where libraries permit.
- Record known nondeterministic operations.
- Store immutable dataset snapshots or retrieval scripts with hashes.
- Publish raw, tidy, and derived results.
- Generate every figure and table through scripts.
- Add `CITATION.cff`, license, code of conduct, and contribution guide.

- Archive a release and artifacts on Zenodo or an equivalent DOI-granting repository.
- Create an anonymized repository snapshot if double-blind review requires it.
- Pre-register research questions, hypotheses, primary metrics, exclusions, and statistical tests before the final benchmark.
- Label post-hoc analyses as exploratory.

19. Human researcher and AI-agent roles



Saranraj U is the researcher, builder, decision-maker, and accountable author. The AI agent is an assistant, not an author, principal investigator, or source of scientific accountability.

Responsibility matrix

Activity	Saranraj U	AI agent
Define research purpose	Own and approve	Suggest alternatives and critique scope
Approve novelty claim	Own after reading literature	Search, summarize, identify competing work
Select datasets and licenses	Decide and verify	Prepare candidate matrix and flag risks
Freeze protocol and hypotheses	Approve and sign off	Draft protocol and consistency-check it
Design architecture	Approve trade-offs	Propose contracts, modules, and tests
Write implementation	Review, run, and understand all code	Generate drafts, tests, refactors, and explanations
Execute experiments	Run or supervise	Prepare commands, monitor outputs when invoked, diagnose failures
Validate results	Independently inspect	Compute checks, summaries, and alternative analyses
Statistical decisions	Approve and justify	Suggest tests, write analysis code, check assumptions
Interpret findings	Own final interpretation	Produce competing interpretations and limitation checks
Draft manuscript	Revise and approve every claim	Draft sections from verified artifacts
Verify citations	Open and verify each cited source	Discover sources and format references
Authorship and submission	Solely human responsibility	No authorship; assist with formatting/checklists
Respond to reviewers	Decide and approve responses	Draft response matrix and analyses
Research integrity	Fully accountable	Flag inconsistencies and unsupported claims

Tasks the AI agent may perform

- Literature discovery and structured summaries
- Drafting research plans, code skeletons, tests, documentation, and manuscript text
- Generating alternative hypotheses and identifying confounders
- Writing reproducible analysis scripts
- Producing figures and tables from verified data
- Checking manuscript consistency against generated artifacts
- Preparing reviewer-response drafts

Tasks the AI agent must not own

- Claiming authorship

- Inventing citations, data, results, experiments, approvals, or participant information
- Making final inclusion/exclusion decisions without human approval
- Submitting a paper or agreeing to publisher terms
- Certifying that generated code or statistics are correct without execution and verification
- Deciding that a finding is scientifically valid merely because output looks plausible

Human verification checklist for AI-generated work

- ☐ I understand the generated code before relying on it.
- ☐ I executed tests and inspected failures.
- ☐ I verified every numerical claim against frozen artifacts.
- ☐ I opened and checked every cited paper.
- ☐ I checked that quotations and paraphrases are accurate.
- ☐ I reviewed assumptions of every statistical test.
- ☐ I documented deviations from the protocol.
- ☐ I ensured no confidential or restricted data were entered into an AI system.
- ☐ I reviewed the target venue's current AI policy.
- ☐ I accept full accountability for the submitted manuscript.

20. AI-use disclosure and audit log

COPE states that AI tools cannot be listed as authors because they cannot take responsibility for submitted work. Publisher policies vary and may require disclosure of the tool, version, purpose, and manner of use. Before submission, read the exact policy of the target venue.

Maintain an AI assistance log

Timestamp	Tool and version	Task	Inputs/data classification	Output used?	Human verification	Commit/section
YYYY-MM-DD	Notion AI / model version if available	Protocol drafting	Public literature and project concept	Yes/No	Description	Link or commit

Draft disclosure statement

Declaration of generative AI-assisted technologies: During the preparation of this work, the author used Notion AI/IGRIS [insert model/version if available] to assist with literature discovery, protocol drafting, software scaffolding, code review, documentation, and language refinement. The author independently reviewed relevant sources, executed and validated all code and experiments, verified all reported results and citations, revised the generated material, and accepts full responsibility for the content of the publication. No AI system is listed as an author.

Adapt this statement to the destination venue. If the venue requires prompt disclosure, export the relevant AI-assistance log as supplementary material.

21. Ethics and responsible-research considerations

- Use only datasets with appropriate licenses and documented provenance.
- Avoid presenting benchmark populations as representative of real communities without evidence.
- Do not make high-stakes deployment recommendations from benchmark datasets.

- Treat demographic attributes carefully and explain why subgroup evaluation is scientifically necessary.
 - Avoid causal language: attribution explains model behavior under an operational definition, not the real-world cause of an outcome.
 - Document environmental and compute costs.
 - Disclose failed methods and negative results.
 - Do not optimize metric definitions after viewing final rankings without labelling the change exploratory.
 - If a future human-subject study is added, obtain the required ethics review before recruitment or data collection.
-

22. Build and research roadmap

Phase 0 — Literature and protocol freeze

Researcher

- Read the foundational and competing papers.
- Verify the novelty matrix.
- Select a target venue family.
- Approve RQs, hypotheses, metrics, and exclusions.
- Freeze Protocol v1.0 before confirmatory experiments.

AI agent

- Maintain a literature matrix.
- Compare ExplainCheck against Quantus, OpenXAI, and other benchmarks.
- Draft preregistration and identify ambiguous definitions.

Exit criteria

- Signed protocol
- Verified bibliography
- Dataset candidate matrix
- Novelty statement that survives comparison to prior work

Phase 1 — Research skeleton

- Typed contracts
- Configuration schema
- Provenance and hashing
- Synthetic generator
- Logistic baseline
- One explainer and one metric
- End-to-end smoke run

Exit criteria: one command generates raw results, JSON, a model card, and a minimal paper table.

Phase 2 — Metric validation

- Implement fidelity, stability, sparsity, and runtime
- Hand-calculated fixtures

- Property tests
- Golden synthetic tests
- Compare selected outputs against an established toolkit where definitions overlap

Exit criteria: every metric has a validated implementation and written limitations.

Phase 3 — Stressors

- Correlation injection
- MCAR and MAR generators
- Imputation strategies
- Prediction-preservation filtering
- Group-attribution analysis

Exit criteria: stress tests reproduce expected behavior on synthetic ground truth.

Phase 4 — Pilot study

- One synthetic and one real dataset
- All primary model/explainer combinations
- Runtime and variance estimation
- Revise compute budget and sample counts without examining final benchmark claims

Exit criteria: final experiment matrix and power/precision rationale are frozen.

Phase 5 — Confirmatory benchmark

- Execute immutable manifests
- Preserve raw results and failures
- Generate statistical outputs
- No unlogged protocol changes

Exit criteria: all primary runs complete and artifact hashes are frozen.

Phase 6 — Analysis and manuscript

- Generate tables and figures
- Write results from generated snippets
- Test hypotheses
- Perform prespecified sensitivity analyses
- Write limitations before finalizing conclusions

Phase 7 — Reproduction and submission

- Clean-environment reproduction
 - Archive release and DOI
 - Venue-specific checklist
 - AI disclosure
 - Internal adversarial review
 - Submission package
-

23. Publication strategy

Paper contribution types

1. **Benchmark/dataset-and-benchmark paper:** strongest if the protocol, datasets, metrics, and empirical findings are substantial.
2. **Research software paper:** appropriate if engineering quality is the principal contribution; consider JOSS-style venues, but do not assume a software paper alone validates empirical novelty.
3. **Workshop or short paper:** suitable for an early validated prototype and focused missingness/correlation result.
4. **Full conference/journal paper:** requires a strong novelty argument, broad experiments, rigorous statistics, and comparison to established frameworks.

Candidate venue families to investigate

- Explainable AI workshops associated with major ML conferences
- ACM FAccT, if the subgroup and governance contribution is deep and socio-technical claims are justified
- AAAI/ACM AIES, if responsible-AI evaluation is central
- Applied AI or machine-learning journals that accept benchmark/tool papers
- JMLR Machine Learning Open Source Software, depending on current scope and requirements
- Journal of Open Source Software for the software artifact, potentially alongside a separate empirical paper

Do not select a venue only by impact factor. Match the contribution, length, artifact policy, review model, and AI-use policy.

Minimum publishability bar

- Clear novelty relative to Quantus and OpenXAI
- At least one non-obvious empirical finding
- Synthetic ground-truth validation
- Multiple real datasets and model families
- Statistical uncertainty and corrected comparisons
- Reproducible public code and artifacts
- Honest limitations and negative results
- No unsupported causal, fairness, or universality claims

24. Manuscript structure

1. Introduction

- Explain why visual plausibility is inadequate.
- State the practical consequences of unreliable explanations.
- Identify the gap in correlation/missingness stress-testing and deployment-ready reporting.
- End with a numbered contribution list.

Copy-ready contribution template

This work makes four contributions. First, we introduce a reproducible protocol for evaluating tabular feature-attribution methods under controlled correlation and missingness stressors. Second, we separate explanation instability from changes in model predictions through prediction-preserving evaluation. Third,

we provide uncertainty-aware and subgroup-aware comparisons across datasets, models, and explainers. Fourth, we release an open-source implementation that generates versioned model cards, machine-readable benchmark artifacts, and CI/CD quality gates from the same frozen experiment record.

2. Related work

- Feature-attribution methods
- XAI evaluation metrics
- XAI benchmark toolkits
- Robustness under correlation and missingness
- Model cards and dataset documentation
- Reproducibility and ML governance

3. Methods

- Study design
- Datasets and datasheets
- Models and training
- Explainers and parameters
- Metric definitions
- Stressor generation
- Statistical analysis
- Reproducibility

4. Results

- Predictive-model performance
- Core metric comparison
- Metric disagreement
- Correlation stress test
- Missingness stress test
- Subgroup analysis
- Runtime and failure analysis
- CI regression case study

5. Discussion

- Principal findings
- Why rankings differ
- Implications for selecting explainers
- Implications for model governance
- Comparison with prior benchmarks
- Limitations
- Future work

6. Conclusion

Answer only the research questions supported by the evidence.

Supplementary material

- Full parameters
 - Additional datasets
 - Complete result tables
 - Sensitivity analyses
 - Failure logs
 - Datasheets
 - Model cards
 - AI assistance disclosure/log if required
-

25. Planned tables and figures

Main tables

- **Table 1:** Dataset characteristics, subgroup variables, missingness, license, and split sizes
- **Table 2:** Predictive performance and calibration by model and dataset
- **Table 3:** Explanation-quality metrics by explainer with 95% confidence intervals
- **Table 4:** Effect of correlation and missingness stressors
- **Table 5:** Subgroup consistency and effect sizes

Main figures

- **Figure 1:** Study and artifact-generation pipeline
- **Figure 2:** Metric trade-off/rank profile without a composite score
- **Figure 3:** Attribution robustness as correlation increases
- **Figure 4:** Explanation drift by missingness rate and handling strategy
- **Figure 5:** Subgroup metric distributions
- **Figure 6:** CI explanation-regression case study

Caption standard

Every caption must state:

- metric direction;
 - sample size;
 - uncertainty representation;
 - aggregation level;
 - stress condition;
 - whether prediction preservation was required.
-

26. Risk register

Risk	Consequence	Mitigation
Novelty overlaps OpenXAI/Quantus	Rejection	Build a direct comparison matrix and emphasize validated stressors, uncertainty, and CI integration
Metrics disagree or are parameter-sensitive	Ambiguous conclusions	Treat disagreement as a result; run prespecified sensitivity analyses

Risk	Consequence	Mitigation
Perturbations create out-of-distribution samples	Invalid fidelity/stability claims	Use constrained/data-aware perturbations and report OOD diagnostics
Correlation redistributes valid group attribution	False robustness failure	Report both individual and grouped attribution
Missingness mechanism is unrealistic	Weak external validity	Separate MCAR/MAR/MNAR, state assumptions, and avoid universal claims
Excessive compute	Incomplete study	Pilot first; freeze a feasible matrix; record efficiency
Dataset ethics or licensing issue	Cannot publish artifacts	Verify before use and maintain datasheets
AI-generated citation or code error	Integrity failure	Human source verification, tests, and AI-use log
Post-hoc hypothesis changes	Inflated evidence	Freeze protocol and label exploratory analyses
Single-number leaderboard misleads users	Oversimplification	Publish metric profiles and Pareto trade-offs

27. Definition of done

Scientific

- ☐ RQs and hypotheses frozen before confirmatory experiments
- ☐ Novelty matrix completed against major competing frameworks
- ☐ Synthetic ground-truth validation passed
- ☐ Real-world experiment matrix completed
- ☐ Confidence intervals and effect sizes reported
- ☐ Multiplicity controlled
- ☐ Limitations and negative findings documented

Engineering

- ☐ One-command reproduction from a clean environment
- ☐ Typed, documented plugin contracts
- ☐ Unit, property, integration, and golden tests pass
- ☐ Every run has immutable provenance
- ☐ Raw and derived artifacts are separated
- ☐ JSON schema is versioned
- ☐ Model card is generated from frozen results
- ☐ CI regression demonstration works

Publication

- ☐ All manuscript numbers generated from artifacts
- ☐ Every citation manually verified
- ☐ Dataset and model cards included
- ☐ Code and artifact release archived with DOI
- ☐ AI-use statement matches the venue policy
- ☐ Reproducibility checklist completed

28. Literature matrix starter

Work	Relevance	ExplainCheck relationship
Ribeiro, Singh, and Guestrin — LIME	Local model-agnostic explanations	Benchmark target; parameters and local fidelity must be explicit
Lundberg and Lee — SHAP	Additive feature-attribution framework	Benchmark target; TreeSHAP and KernelSHAP treated separately
Mitchell et al. — Model Cards	Structured model reporting	ExplainCheck auto-populates evidence-backed explanation sections
Gebru et al. — Datasheets for Datasets	Dataset documentation	Every benchmark dataset receives a datasheet
Hedström et al. — Quantus	Quantitative evaluation toolkit	Competing/foundational toolkit; compare overlapping metrics
Agarwal et al. — OpenXAI	Benchmark for post-hoc explanations	Closest competing benchmark; novelty must be explicit
NIST AI RMF 1.0	Trustworthy-AI risk framework	Motivates governance and measurement, not a validation claim
COPE AI authorship guidance	Publication ethics	AI agent disclosed as assistance, never author

29. Verified starting references

1. Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *KDD 2016*.
2. Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS 2017*.
3. Mitchell, M., et al. (2019). Model Cards for Model Reporting. *FAT 2019**.
4. Gebru, T., et al. (2021). Datasheets for Datasets. *Communications of the ACM*, 64(12), 86–92.
5. Hedström, A., et al. (2023). Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond. *JMLR*, 24(34), 1–11.
6. Agarwal, C., et al. (2022). OpenXAI: Towards a Transparent Evaluation of Post hoc Model Explanations. *NeurIPS Datasets and Benchmarks*.
7. NIST. (2023). Artificial Intelligence Risk Management Framework 1.0.
8. COPE. Authorship and AI tools.
9. Gundersen, O. E., et al. (2024). Reproducibility in Machine Learning-based Research: Overview, Barriers and Drivers.
10. Collins, G. S., et al. (2024). TRIPOD+AI statement. *BMJ*, 385.



This is a verified **starting bibliography**, not the final systematic review. Before claiming novelty, search forward and backward citations, read the full papers, record inclusion decisions, and update the literature matrix.

30. Immediate next actions

1. Read Quantus and OpenXAI completely and build a feature/metric/claim comparison table.
2. Decide whether the primary paper is a benchmark paper, software paper, or paired artifact strategy.

3. Select candidate datasets only after license and datasheet review.
4. Implement the synthetic linear ground-truth case first.
5. Validate one fidelity and one stability metric by hand.
6. Produce one end-to-end artifact bundle before adding more methods.
7. Conduct a compute/variance pilot.
8. Freeze Protocol v1.0 and preregister before the confirmatory run.



Governing principle: Build evidence first, software second, and presentation third. Every code path should support a research question, every reported number should be reproducible, and every scientific claim remains Saranraj U's responsibility.



ExplainCheck — Phase 0 Execution, Pilot & Preregistration Package